Analytical Cumulative examination

Friday 21st January 2000

(1) Calculate the energy (in kJ) liberated in the fusion reaction to produce one mole of helium from deuterium:

$$^{2}H + {}^{2}H \rightarrow {}^{4}He$$

(2) How many gallons of gasoline (octane) having a density 0.703 g/mL, would have to be burned to produce the same amount of energy as in the production of 1 mol of ^{4}He by fusion of deuterium.

(3) Calculate the binding energy in kJ/mol and in MeV/nucleon for ^{2}H, ^{4}He, ^{7}Li, ^{14}N and ^{19}F and discuss (and where possible, explain) their relative values.

(4) Discuss the value of this reaction as an energy source and the current status of efforts to use the reaction commercially.

Table of atomic masses (u) and constants
p = 1.007277
n = 1.008665
e^{-} = 5.4859 x 10^{-4}
^{2}H = 2.014102
^{4}He = 4.002603
^{7}Li = 7.01600
^{4}N = 14.003074
^{4}F = 18.99840
c = 2.9979 x 10^{8} m/s
1 MeV = 9.65 x 10^{7} kJ/mol